

Ethan Cao

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EDUCATION & CERTIFICATIONS

University of California, Santa Cruz *B.S. Computer Science*

Expected Graduation June 2027

GPA: 3.87/4.0 GPA, Dean's Honors List

Relevant Courses: Data Structures and Algorithms, Algorithmic Foundations of Cryptography, Analysis of Algorithms, Computer Architecture, Computer Systems and C Programming, Computer Systems and Assembly Language

CompTIA Network+ (N10-009), **CompTIA** Security+ (SY0-701)

EXPERIENCE

Nintendo, *Incoming Security Engineering Intern*

Redmond, WA | June 2026 – August 2026

Farm-ng (Bonsai Robotics), *Software Engineer Intern*

Watsonville, CA | June 2025 – August 2025

- Designed and deployed a secure user management system, integrated into over 150 production robots.
- Implemented multi-factor authentication using OAuth2, passwords, and hardware-bound cryptographic keys.
- Developed encrypted gRPC/Protobuf protocols to secure robot–Fleet Manager communications.
- Built a full-stack interface (React + FastAPI) enabling authenticated field operations on robotic devices.
- Collaborated with robotics engineers to debug and troubleshoot production units, resolving software–hardware integration issues and reducing field downtime.

RANDLab @ UCSC, *Undergraduate Security Researcher*

Santa Cruz, CA | January 2026 – Present

- Collaborated with Prof. Ram Sundara Raman at RANDLab (Research on Attacks on Networks and Data) to measure geographic variation in web accessibility, privacy disclosures, and tracking behavior.
- Integrated Bright Data proxies into GeoInspector and privacy-policy scraping for cross-region DNS/TLS/HTTP and policy comparison, successfully parsing through the top 1,000 Tranco-listed sites so far.
- Ran layered connectivity experiments across DNS, TCP/TLS, and HTTP to localize whether cross-region differences stem from resolution, transport, or application-layer behavior.
- Advanced browser-level measurement (cookies, third-party requests, consent flows) to surface tracking and disclosure differences that reachability-only tests miss.

IT Consultant Services (Ichor Systems), *System and Security Specialist*

San Jose, CA | February 2021 – Present

- Administered Red Hat and Oracle Linux, VMware, Windows, data backups, and LDAP, and configured firewalls, TCP Wrappers, and remote access controls to reduce unauthorized access attempts.
- Managed 15+ Linux servers across development, staging, and production, maintaining high availability while hardening systems and applying security patches within 48 hours of release.
- Supported data-center operations with real-time monitoring and alerting for faster incident response and fewer service disruptions.

UCSC Information Technology Services, *IT Consultant*

Santa Cruz, CA | May 2024 – Present

- Provided front-line help-desk support for campus Learning Technologies, triaging walk-ups and tickets for printers, MacBooks, AV, and lab equipment during peak class periods.
- Collaborated with roughly 25 peer consultants and 3 student managers on scheduling, coverage, and handoffs.

PROJECTS

LockIT, *Swift, Cryptography, Apple Security Frameworks*

- Built a macOS app in Swift/SwiftUI to encrypt and lock folders using AES-256-GCM and Touch ID authentication.
- Integrated Apple Security frameworks (Keychain Services, LocalAuthentication, App Sandbox) to protect keys and enforce biometric-gated access.
- Engineered secure archive and decryption workflows with ZIPFoundation, SHA256-derived key recovery, and automatic cleanup.

Sentinel-AntiCheat, *Anti-cheat, Java, Wazuh (SIEM), Bukkit API*

- Developed a real-time anti-cheat engine (Java 21, Paper 1.21) for large multiplayer servers using mining-depth, block-ratio, and behavioral signals.
- Built a Wazuh SIEM pipeline with structured JSONL, custom decoders and rules, and dashboards for alert triage and detection metrics.

AWS Three-Tier Cloud Architecture (Lab), *AWS VPC, EC2, ALB, RDS PostgreSQL, IAM, CloudWatch*

- Built a three-tier AWS lab with a VPC across public/private subnets, Internet and NAT gateways, an Application Load Balancer, private EC2 web servers, and private RDS PostgreSQL.
- Restricted tier-to-tier access with security groups (ALB to EC2, EC2 to RDS on 5432), validated EC2-to-RDS connectivity with psql, and enabled CloudWatch monitoring for EC2 and RDS.

TECHNICAL SKILLS

Languages & frameworks: Python, C, C++, TypeScript, Swift/SwiftUI, React, FastAPI, gRPC/Protobuf

Security, systems & cloud: Linux, VMware, Cryptography, Wazuh SIEM, Security Monitoring & Alerting, Threat Detection, AWS

Spoken languages: English, Vietnamese